IoT Challenge 3

Exercise Report – Part 2

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1 Introduction

This document contains the solution to Part 2 of IoT Challenge 3, which concerns the dimensioning and analysis of a LoRaWAN network for IoT applications. The exercise considers a network deployed in Europe at 868 MHz composed of a single gateway and 40 sensor nodes, each transmitting 20-byte payloads at a rate of $\lambda = 2$ packets per minute according to a Poisson process.

The objective is twofold. First, EQ1 asks to determine the largest spreading factor that still guarantees a packet success rate of at least 75%, using a simplified uncoordinated transmission model and the TTN airtime calculator to obtain the on-air duration of each packet. Second, EQ2 presents a real-world deployment scenario where the observed performance is lower than expected and non-uniform across nodes, and asks which corrective action is most appropriate.

2 Scenario Description

The LoRaWAN network under analysis has the following fixed parameters throughout the exercise:

- Carrier frequency: **868 MHz** (EU band)
- Bandwidth: **125 kHz**
- Number of gateways: **1**
- Number of sensor nodes: **40**
- Payload size: **20 bytes**
- Transmission rate per node: $\boldsymbol{\lambda = 2}$ **packets/minute** (Poisson process)
- All nodes use the **same spreading factor**
- Transmissions are **uncoordinated** and may overlap in time

The packet success rate is modeled under the assumption that two overlapping transmissions at the same spreading factor result in both packets being lost (collision). The airtime of each packet is computed using the TTN airtime calculator at https://www.thethingsnetwork.org/airtime-calculator.

3 EQ1 – Largest SF Guaranteeing a 75% Packet Success Rate

3.1 Model

Under the simplified uncoordinated model, each node transmits according to a Poisson process with rate λ. The total aggregate traffic generated by all $N = 40$ nodes is therefore a Poisson process with rate:

$$\Lambda = N \cdot \lambda = 40 \times 2 = 80 \text{ packets/minute}$$

For a given spreading factor, let T_{air} be the on-air duration of a single packet (obtained from the TTN calculator). The probability that a given packet is received successfully is the probability that no other transmission starts during its vulnerable period of duration $2\,T_{air}$ (since a collision occurs if any other packet overlaps with it). Under the Poisson assumption, this is:

$$P_{success} = e^{-\Lambda' \cdot 2\,T_{air}}$$

where Λ' is the aggregate rate of *all other* nodes, i.e. $\Lambda' = (N-1) \cdot \lambda = 39 \times 2 = 78$ packets/minute, or equivalently the total rate Λ is used as an approximation when N is large. The goal is to find the largest SF such that $P_{success} \geq 0.75$.

3.2 Airtime Computation

Spreading Factor	**Airtime T_{air} (ms)**	$P_{success}$
SF7	71.6	0.826
SF8	133.6	0.700
SF9	246.8	< 0.75
SF10	452.6	< 0.75
SF11	987.1	< 0.75
SF12	1810.4	< 0.75

Table 1: *On-air time and packet success rate for each spreading factor.*

3.3 Result

The packet success rate is computed using the pure ALOHA model:

$$P_{success} = e^{-2G}, \quad G = N \cdot \lambda \cdot T_{air}$$

where $N = 40$ nodes, $\lambda = 2$ packets/minute, and T_{air} is expressed in minutes.

SF7:

$$G = 40 \cdot \frac{2}{60} \cdot \frac{71.6}{1000} \cdot 60 = 0.0955, \quad P_{success} = e^{-2 \cdot 0.0955} = e^{-0.1909} \approx 0.826 \geq 0.75$$

SF8:

$$G = 40 \cdot \frac{2}{60} \cdot \frac{133.6}{1000} \cdot 60 = 0.1781, \quad P_{success} = e^{-2 \cdot 0.1781} = e^{-0.3563} \approx 0.700 < 0.75 \times$$

EQ1 answer: The largest spreading factor that guarantees a packet success rate of at least 75% is **SF 7**.

4 EQ2 – Corrective Action for Non-Uniform Performance

4.1 Problem Description

After deploying the network in the field, two anomalies are observed with respect to the theoretical model. First, the overall measured packet success rate is significantly lower than the value predicted in EQ1. Second, the performance is non-uniform across nodes: some nodes achieve relatively good delivery rates, while others experience very poor performance.

The three candidate corrective actions proposed are:

1. Decrease the number of nodes
2. Move the nodes closer to the gateway
3. Increase the spreading factor

4.2 Analysis of Each Option

In a LoRaWAN network, a packet is successfully received only if two independent conditions are simultaneously satisfied: (i) the received signal power at the gateway exceeds the sensitivity threshold ($P_{RX} > S_{RX}$), and (ii) no collision occurs with another transmission on the same frequency and spreading factor. These two failure modes have fundamentally different spatial signatures, which is the key to interpreting the observed symptoms.

Option 1 – Decrease the number of nodes. Reducing the number of nodes lowers the aggregate channel load $G = N\lambda T_{air}$, which directly decreases the collision probability according to the ALOHA model:

$$P_{success} = e^{-2G} = e^{-2N\lambda T_{air}}$$

This action would improve the packet success rate for all nodes uniformly, since under a Poisson traffic model every node is statistically equivalent with respect to collisions. However, it does not explain nor resolve the observed non-uniformity: nodes that currently suffer from poor performance would continue to do so, as their problem is not related to channel congestion.

Option 2 – Move the nodes closer to the gateway. In a real deployment, each node experiences a different path loss depending on its distance from the gateway and on the physical obstacles in its surroundings. A packet can be decoded only if the received power exceeds the gateway sensitivity, which depends on the spreading factor and bandwidth:

$$P_{min} = -174 + 10\log_{10}(BW) + NF + SNR$$

Nodes that are far away or obstructed may operate at or below this threshold, causing systematic and persistent packet loss that is independent of traffic load. Since each node has a distinct physical location, heterogeneous path loss naturally produces non-uniform delivery rates across nodes – exactly the symptom described in the problem. Moving nodes closer to the gateway increases the received signal power for all nodes, and disproportionately benefits those that were previously operating near or below the sensitivity floor, thereby both improving the overall success rate and reducing the performance gap between nodes.

Option 3 – Increase the spreading factor. A higher SF lowers the receiver sensitivity threshold, allowing the gateway to decode weaker signals and potentially recovering some of the nodes suffering from high path loss. However, a higher SF also increases the packet airtime significantly (airtime grows exponentially with SF), which raises the aggregate channel load G and therefore increases the collision probability for all nodes. The two effects partially cancel each other: coverage improves but collision performance degrades. This option does not directly address the root cause of the non-uniformity and introduces an adverse side effect on network capacity.

4.3 Chosen Action and Justification

The key diagnostic clue in this problem is the *non-uniformity* of performance across nodes. Under any collision-based model, all nodes share the same channel and experience statistically similar collision probabilities; therefore, collisions alone cannot produce the observed heterogeneous behavior. Non-uniform performance is instead the direct signature of heterogeneous path loss: nodes at different distances or with different obstructions reach the gateway with different received powers, causing some to fall below the sensitivity threshold while others remain well above it.

Option 2 directly addresses this root cause by closing the gap between each node's received power and the gateway sensitivity threshold. It improves the success rate of the weakest nodes without worsening the collision probability, making it the most targeted and effective corrective action.

EQ2 answer: The most appropriate action is **Option 2 – Move the nodes closer to the gateway**.

The non-uniform performance across nodes is the signature of heterogeneous path loss in a real environment: nodes that are farther from or obstructed from the gateway receive a signal below the sensitivity threshold, causing systematic losses that affect some nodes far more than others. Moving nodes closer directly increases the received signal power, addressing the root cause without worsening the collision probability.

5 Conclusion

This exercise highlights two fundamental trade-offs in the design and deployment of LoRaWAN networks. For EQ1, the results show that increasing the spreading factor improves receiver sensitivity and link budget, but comes at the cost of longer packet airtimes; under the ALOHA collision model, longer airtimes raise the probability of overlap between concurrent transmissions, eventually driving the packet success rate below the 75% threshold for sufficiently high SF values in a dense network of 40 nodes. For EQ2, the non-uniform delivery performance observed across nodes is a strong diagnostic indicator that the dominant failure mode in the real deployment is not channel congestion but inadequate received signal power for a subset of nodes – a consequence of heterogeneous path loss that a theoretical model with no spatial dimension cannot capture. Among the three candidate actions, moving the nodes closer to the gateway is the most targeted solution because it directly increases the received signal power for the weakest nodes without the adverse side effect on collision probability that an SF increase would introduce.